

# Plan

- Similar Matrices
- Eigenvalues and Eigenvectors
- Diagonalizable Matrix

# Similar Matrices

## Definition

Let  $A$  and  $B$  be two  $n \times n$  matrices. Then  $A$  is said to be similar to  $B$  if there is an  $n \times n$  invertible matrix  $P$  such that

$A = P^{-1}BP$ . We then write  $A \approx B$ .

- If  $A \approx B$ , we can equivalently write that  $B = PAP^{-1}$  or  $BP = PA$ .
- (i)  $A \approx A$ , (ii)  $A \approx B \implies B \approx A$ ,  
(iii)  $A \approx B$ , and  $B \approx C \implies A \approx C$ .
- Suppose  $A = P^{-1}BP$ . Then for any  $k \in \mathbb{N}$ ,  $A^k = P^{-1}B^kP$ .  
In particular, if  $B$  is a diagonal matrix, then computing  $A^k$  is easy!

## Definition

An  $n \times n$  matrix is said to be diagonalizable if it is similar to a diagonal matrix.

## Result

Let  $A, B$  be similar matrices ( $A = P^{-1}BP$ ). Then

- 1  $\det A = \det B$ .
- 2  $A$  is invertible *iff*  $B$  is invertible.
- 3  $A$  and  $B$  have the *same rank*.
- 4  $A^m \approx B^m$  for all integers  $m \geq 0$ .
- 5 If  $A$  is invertible, then  $A^{-1} \approx B^{-1}$ .
- 6 If  $A$  is invertible, then  $A^m \approx B^m$  for *all integers*  $m$ .

### Example

Let  $A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$ ,  $B = \begin{bmatrix} 1 & 0 \\ -2 & -1 \end{bmatrix}$  and  $T = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ .

Then  $A \approx B$ , since  $AT = TB = \begin{bmatrix} 3 & 1 \\ -1 & -1 \end{bmatrix}$ .

### Exercise

Is  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$  similar to  $\begin{bmatrix} d & c \\ b & a \end{bmatrix}$ ? Take  $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ .

In general, it is not easy to find out whether two matrices are similar.

**Note:** Suppose  $AP = PD$  where  $P = [\mathbf{x}_1, \dots, \mathbf{x}_n]$  and  $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ . Then

$$[A\mathbf{x}_1, \dots, A\mathbf{x}_n] = AP = PD = [\lambda_1\mathbf{x}_1, \dots, \lambda_n\mathbf{x}_n],$$

i.e.  $A\mathbf{x}_j = \lambda_j\mathbf{x}_j$ ,  $1 \leq j \leq n$ .

# Eigenvalues and Eigenvectors

## Definition

Let  $A$  be an  $n \times n$  matrix.

- A complex number  $\lambda$  is called an eigenvalue of  $A$  if there is  $\mathbf{x} \in \mathbb{C}^n$ ,  $\mathbf{x} \neq \mathbf{0}$  such that  $A\mathbf{x} = \lambda\mathbf{x}$ .
- We call  $\mathbf{x}$  an eigenvector of  $A$  corresponding to  $\lambda$ .

## Example

For  $\begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$ , eigenvalues:  $4, -2$ ; resp. eigenvectors  $\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ .

## Result

Let  $A$  be an  $n \times n$  matrix. Then

- $\lambda$  is an eigenvalue of  $A$  iff  $A - \lambda I$  is not invertible.
- $\lambda$  is an eigenvalue of  $A$  iff  $\det(A - \lambda I) = 0$ .
- $0$  is an eigenvalue of  $A$  iff  $A$  is not invertible.

## Definition

Let  $\lambda$  be an **eigenvalue** of a matrix  $A$ . The **eigenspace of  $\lambda$**  is defined to be

$$E_\lambda = \{\mathbf{x} \in \mathbb{C}^n : A\mathbf{x} = \lambda\mathbf{x}\}.$$

- The **nonzero vectors** in  $E_\lambda$  are **eigenvectors** of  $A$  corresponding to  $\lambda$ .
- $E_\lambda = \text{null}(A - \lambda I)$ , and is a nontrivial **subspace** of  $\mathbb{C}^n$ .
- Let  $\mathbf{v}_1, \dots, \mathbf{v}_k$  be **eigenvectors** of  $A$  corresponding to  $\lambda$  and  $\mathbf{v} = \alpha_1\mathbf{v}_1 + \dots + \alpha_k\mathbf{v}_k \neq \mathbf{0}$ . Then  $\mathbf{v}$  is an eigenvector of  $A$  corresponding to  $\lambda$ .

## Definition

Let  $A$  be an  $n \times n$  matrix. Then the **characteristic polynomial** of  $A$  is defined to be  $P_A(x) = \det(A - xI)$ .

The equation  $P_A(x) = 0$  is called **characteristic equation** of  $A$ .

**Note:**  $\lambda$  is an eigenvalue of  $A$  iff  $P_A(\lambda) = 0$ .

## Method of Finding Eigenvalues $\lambda$ and Bases for $E_\lambda$

- 1 Compute the characteristic polynomial  $\det(A - xI)$ .
- 2 Solve the characteristic equation  $\det(A - xI) = 0$  for eigenvalues  $x = \lambda$ .
- 3 For each eigenvalue  $\lambda$ , solve the system  $(A - \lambda I)\mathbf{x} = \mathbf{0}$  and find a basis for the eigenspace  $E_\lambda$ .

### Example

Take  $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} -1 & 0 & 1 \\ 3 & 0 & -3 \\ 1 & 0 & -1 \end{bmatrix}$ .

- $P_A(x) = -(x-1)^2(x-2)$ ;  $E_1 = \text{span}\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right)$ ,  $E_2 = \text{span}\left(\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}\right)$ .
- $P_B(x) = -x^2(x+2)$ ;  $E_0 = \text{span}\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right)$ ,  $E_{-2} = \text{span}\left(\begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix}\right)$ .

## Result

- Eigenvalues of a *triangular* matrix are its *diagonal entries*.
- Eigenvalues of the matrix  $\left[ \begin{array}{c|c} A_p & C \\ \hline O & B_q \end{array} \right]$  are precisely the eigenvalues of  $A_p$  and  $B_q$ .

## Example

*A real matrix may have complex eigenvalues and complex eigenvectors. Consider  $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ .*

$P_A(x) = x^2 + 1$ , eigenvalues are  $\pm i$ , with corresponding eigenvectors  $[1, i]^T$  and  $[i, 1]^T$ .

The eigenspaces  $E_i$  and  $E_{(-i)}$  are one dimensional.



## Result

Let  $A$  be a matrix, and  $\lambda$  is an eigenvalue of  $A$  with corresponding eigenvector  $\mathbf{x}$ . Then  $A\mathbf{x} = \lambda\mathbf{x}$ .

- 1 For  $n \in \mathbb{N}$ ,  $\lambda^n$  is an eigenvalue of  $A^n$  with corresponding eigenvector  $\mathbf{x}$ .
- 2 If  $A$  is invertible, then  $\frac{1}{\lambda}$  is an eigenvalue of  $A^{-1}$  with corresponding eigenvector  $\mathbf{x}$ .

## Result

Let  $A, B$  be similar matrices ( $A = P^{-1}BP$ ). Then

- 1  $A$  and  $B$  have the same characteristic polynomial.
- 2  $A$  and  $B$  have the same set of eigenvalues.
- 3 Note:  $BP = PA$  gives  $A\mathbf{v} = \lambda\mathbf{v} \iff B(P\mathbf{v}) = \lambda(P\mathbf{v})$ . Thus,  $\lambda$  is an eigenvalue of  $A$  with eigenvector  $\mathbf{v}$  iff  $\lambda$  is an eigenvalue of  $B$  with eigenvector  $P\mathbf{v}$ .
- 4 The  $\dim(E_\lambda)$  for  $A$  is same as  $\dim(E_\lambda)$  for  $B$ .

# Diagonalizable Matrices

**Recall:**  $A$  is **diagonalizable** if there is a diagonal matrix  $D$  such that  $A \approx D$ , i.e.,  $AP = PD$  for some invertible  $P$ .

## Example

The matrix  $A = \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix}$  is diagonalizable:

if  $D = \begin{bmatrix} 4 & 0 \\ 0 & -1 \end{bmatrix}$  and  $P = \begin{bmatrix} 1 & 3 \\ 1 & -2 \end{bmatrix}$  then  $AP = PD$ .

Suppose  $AP = PD$  where  $P = [\mathbf{x}_1, \dots, \mathbf{x}_n]$  and  $D = \text{diag}(\lambda_1, \dots, \lambda_n)$  ( $\lambda_i$  not necessarily distinct). Then

$$[A\mathbf{x}_1, \dots, A\mathbf{x}_n] = AP = PD = [\lambda_1\mathbf{x}_1, \dots, \lambda_n\mathbf{x}_n],$$

i.e.  $A\mathbf{x}_j = \lambda_j\mathbf{x}_j$ ,  $1 \leq j \leq n$ . Thus,

- The **diagonal entries** of  $D$  are eigenvalues of  $A$ .
- The columns  $\mathbf{x}_j$  of  $P$  are **linearly independent** eigenvectors of  $A$  corresponding to the eigenvalues  $\lambda_j$ .

Let  $A$  be an  $n \times n$  matrix. Suppose  $\mathbf{x}_1, \dots, \mathbf{x}_n$  be linearly independent eigenvectors of  $A$  corresponding to eigenvalues  $\lambda_1, \dots, \lambda_n$ . Put  $P = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_n]$ . Then

- $P$  is invertible, and
- $AP = PD$ , i.e.,  $A \approx D$ .

Hence,  $A$  is diagonalizable.

We have proved the following:

### Result

Let  $A$  be an  $n \times n$  matrix. Then  $A$  is diagonalizable iff  $A$  has  $n$  linearly independent eigenvectors.

## Example

Check for the *diagonalizability*. If diagonalizable, find a  $P$  that diagonalizes it. [Use GJE]

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} -1 & 0 & 1 \\ 3 & 0 & -3 \\ 1 & 0 & -1 \end{bmatrix}.$$

- For  $A$ ,  $E_1 = \text{span}\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right)$ ,  $E_2 = \text{span}\left(\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}\right)$ .
- For  $B$ ,  $E_0 = \text{span}\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right)$ ,  $E_{-2} = \text{span}\left(\begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix}\right)$ .
- $A$  is **not** diagonalizable.  $B$  is diagonalizable,  $P = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 3 \\ 0 & 1 & 1 \end{bmatrix}$ .

## Result

Let  $\lambda_1, \lambda_2, \dots, \lambda_m$  be *distinct* eigenvalues of  $A$  with corresponding eigenvectors  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ . Then  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$  are *linearly independent*.

**Proof.** Suppose, *if possible*, for some  $k$ ,  $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$  are linearly independent and  $\mathbf{v}_k = c_1 \mathbf{v}_1 + \dots + c_{k-1} \mathbf{v}_{k-1}$ . Then,

$$A\mathbf{v}_k = \lambda_k \mathbf{v}_k = c_1 \lambda_1 \mathbf{v}_1 + \dots + c_{k-1} \lambda_{k-1} \mathbf{v}_{k-1}.$$

Moreover,

$$\lambda_k \mathbf{v}_k = c_1 \lambda_k \mathbf{v}_1 + \dots + c_{k-1} \lambda_k \mathbf{v}_{k-1}.$$

Therefore,

$$\mathbf{0} = c_1(\lambda_1 - \lambda_k) \mathbf{v}_1 + \dots + c_{k-1}(\lambda_{k-1} - \lambda_k) \mathbf{v}_{k-1},$$

and so,  $c_1(\lambda_1 - \lambda_k) = 0, \dots, c_{k-1}(\lambda_{k-1} - \lambda_k) = 0$ . Since  $c_i \neq 0$  for some  $i$ ,  $\lambda_i = \lambda_k$  for some  $1 \leq i \leq k-1$ , *not possible*.  $\square$

## Result

If  $A$  is an  $n \times n$  matrix with  $n$  distinct eigenvalues then  $A$  is diagonalizable.

## Result

Let  $\lambda_1, \lambda_2, \dots, \lambda_k$  be distinct eigenvalues of a matrix  $A$ . If  $\mathcal{B}_i$  is a basis for the eigenspace  $E_{\lambda_i}$ , then  $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2 \cup \dots \cup \mathcal{B}_k$  is a linearly independent set.

## Definition

Let  $\lambda$  be an eigenvalue of a matrix  $A$ .

- The algebraic multiplicity of  $\lambda$  is the multiplicity of  $\lambda$  as a root of the characteristic polynomial of  $A$ .
- The geometric multiplicity of  $\lambda$  is the dimension of  $E_{\lambda}$ .

## Result

The *geometric multiplicity* of an eigenvalue  $\lambda$  of  $A$  is *less than or equal* to its *algebraic multiplicity*.

**Proof.** Suppose  $A\mathbf{v}_j = \lambda\mathbf{v}_j$ ,  $1 \leq j \leq k$ , where  $\mathbf{v}_1, \dots, \mathbf{v}_k$  are linearly independent. Let  $\mathbf{v}_{k+1}, \dots, \mathbf{v}_n$  be such that

$$P = [\mathbf{v}_1 \cdots \mathbf{v}_k \mid \mathbf{v}_{k+1} \cdots \mathbf{v}_n] = [U \mid V]$$

is invertible. Then  $AU = \lambda U$ . Let  $P^{-1} = \begin{bmatrix} R \\ S \end{bmatrix}$ . Then

$$\left[ \begin{array}{c|c} I_k & \mathbf{0} \\ \hline \mathbf{0} & I_{n-k} \end{array} \right] = I_n = P^{-1}P = \begin{bmatrix} R \\ S \end{bmatrix} [U \mid V] = \left[ \begin{array}{c|c} RU & RV \\ \hline SU & SV \end{array} \right].$$

Thus,  $RU = I_k$ ,  $SV = I_{n-k}$ ,  $RV = SU = \mathbf{0}$ , and

$$P^{-1}AP = \begin{bmatrix} R \\ S \end{bmatrix} A[U \mid V] = \left[ \begin{array}{c|c} RAU & RAV \\ \hline SAU & SAV \end{array} \right] = \left[ \begin{array}{c|c} \lambda RU & RAV \\ \hline \lambda SU & SAV \end{array} \right] = \left[ \begin{array}{c|c} \lambda I_k & RAV \\ \hline \mathbf{0} & SAV \end{array} \right].$$

Thus,  $P_A(x) = \det(P^{-1}AP - xI) = (\lambda - x)^k \det(SAV - xI_{n-k})$ .

So, algebraic multiplicity of  $\lambda$  is **at least  $k$** . □

## Result (The Diagonalization Theorem)

Let  $A$  be an  $n \times n$  matrix whose *distinct eigenvalues* are  $\lambda_1, \lambda_2, \dots, \lambda_k$ . Then the following statements are equivalent:

- 1  $A$  is diagonalizable.
- 2  $A$  has  $n$  linearly independent eigenvectors.
- 3 The union  $\mathcal{B}$  of the bases of the eigenspaces of  $A$  contains  $n$  vectors.
- 4 The *algebraic multiplicity* of each eigenvalue *equals* its *geometric multiplicity*.



### Result (Fundamental Theorem of Invertible Matrices: III)

Let  $A$  be an  $n \times n$  matrix. Then the following statements are equivalent.

1.  $A$  is *invertible*.
2.  $A^t$  is *invertible*.
3.  $A\mathbf{x} = \mathbf{b}$  has a *solution* for every  $\mathbf{b}$  in  $\mathbb{F}^n$ .
4.  $A\mathbf{x} = \mathbf{b}$  has a *unique solution* for every  $\mathbf{b}$  in  $\mathbb{F}^n$ .
5.  $A\mathbf{x} = \mathbf{0}$  has only the *trivial solution*.
6. The reduced row echelon form of  $A$  is  $I_n$ .
7. The rows of  $A$  are linearly independent.
8. The columns of  $A$  are linearly independent.
9.  $\text{rank}(A) = n$ .
10.  $A$  is a *product of* elementary matrices.

- 11.  $\text{nullity}(A) = 0$ .
- 12.  $\text{col}(A) = \mathbb{F}^n$ .
- 13. The column rank of  $A$  is  $n$ .
- 14.  $\text{row}(A) = \mathbb{F}^n$ .
- 15. The row rank of  $A$  is  $n$ .
- 16.  $\det(A) \neq 0$ .
- 17.  $T_A : \mathbb{F}^n \rightarrow \mathbb{F}^n$  is one-one, i.e.,  $\ker(T_A) = \{\mathbf{0}\}$ .
- 18.  $T_A : \mathbb{F}^n \rightarrow \mathbb{F}^n$  is onto, i.e.,  $\text{range}(T_A) = \mathbb{F}^n$ .
- 19.  $T_A : \mathbb{F}^n \rightarrow \mathbb{F}^n$  is an isomorphism.
- 20.  $0$  is not an eigenvalue of  $A$ .

# CAYLEY-HAMILTON THEOREM

Let  $p(\lambda)$  be the characteristic polynomial of a matrix  $A$ . Then  $p(A) = O$ , the zero matrix.

[This is a beautiful and useful theorem. Proof will be supplied later.]

## Example

Let  $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ .

- Verify that characteristic polynomial of  $A$  is  $\lambda^2 - 2\lambda - 3$ .
- Verify that  $A^2 - 2A - 3I = O$ .
- Use the fact  $A^2 = 2A + 3I$  to compute  $A^5$  without computing any matrix multiplication.
- Argue that  $A$  is invertible using its characteristic polynomial. Since  $3I = A^2 - 2A$ , we have  $A^{-1} = \frac{1}{3} [A - 2I]$ ..